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IVS Project – (INTERCALCULATOR)

VERSION 1.0.1  
**Programmer Manual**

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Current as of April 27, 2026

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# 1 Current Appearance of the Calculator



## 2 Introduction

This document serves as a supplementary manual for developers contributing to the INTERCALCULATOR project. Reading the entire manual is strongly recommended. This document is not intended for end users of the application.

### 2.0.1 Getting Started

1) It is recommended to clone the project into a chosen target directory:

```
$ git clone https://github.com/honzakralem/ivs-project-2
```

2) Create a virtual environment using the make command:

```
$ make
```

3) Activate the virtual environment. This will differ depending on the operating system. For Linux/macOS:

```
$ source env/bin/activate
```

For Windows:

```
$ env\bin\activate
```

4) It is strongly recommended to configure your git name and email (although it is not mandatory):

```
$ git config --global user.name "name"
$ git config --global user.email "email"
```

## 3 Prerequisites

For the current phase of development, the following tools/programs must be installed:

- **PYTHON3**: required for running the code itself
- **GNU Make**: required for running automation scripts
- **GIT**: required for version control and collaboration with other developers via the remote repository
- **DOXYGEN**: used for generating documentation

## 4 Architecture

This section covers the logic and structure of the project. Files and directories along with their roles are described here.

- **src** - source code
  - **mathlib.py** - implementation of mathematical functions
  - **mathlib\_test.py** - unit tests that verify the correct functionality of mathlib.py

- `requirements.txt` - list of external libraries required by the project to run or for testing
  - `infixtopost.py` - converts a string from infix notation (e.g.  $2 + 2$ ) to postfix notation (e.g. `2 2 +`)
  - `infixtopost_test.py` - unit tests for the functionality of `infixtopost.py` functions
  - `gui.py` - main graphical user interface
  - `setup.py` - script used to install dependencies
  - `Doxyfile` - configuration file for *Doxygen* documentation
  - The directory also contains `makefile` and `.gitignore`
- `mockup` - directory containing the future user interface design of the calculator
  - `plan` - directory containing the project plan and deadlines
  - `profiling` - directory containing programs for measuring calculator performance
  - The root project directory will contain `makefile`, `LICENSE`, `README.MD`, and `.gitignore`

## 5 Testing

This section covers testing the functionality of modules. To run the tests, simply execute the following command from any directory containing a `makefile`:

```
$ make test
```

This will run `infixtopost_test.py` and `mathlib_test.py`. The result will be reported with a message.

## 6 Makefile and Automation

This section deals primarily with the `makefile` in the root directory of the project. It contains commands to automate development. All commands can be run (assuming the current working directory is the project root) using:

```
$ make help
```